DTS AND COMPUTER CONTROL

Annotations

- 1.a Common, seen similar.
- 1.b Common, seen similar.
- 1.c Uncommon, application of knowledge.
- 1.d Common, application of knowledge.
- 2.a Application of knowledge.
- 2.c Application of knowledge.
- 2.d Application of knowledge.
- 3.a Rare, seen in class.
- 3.b Common, seen similar.
- 3.c Application of knowledge.
- 3.d Rare, seen similar.
- 4.a Seen in class.
- 4.b New.
- 4.c New.
- 4.d New.

Answers

1. a) The equivalent discrete-time model is

$$\begin{aligned} HP(z) &= (1-z^{-1})Z\left(\frac{2}{s(s-2)}\right) \\ &= \frac{e^{0.08}-1}{z-e^{0.08}} = \frac{0.0833}{z-1.0833}. \end{aligned}$$

[4 marks]

b) The transfer function in the w-plane is (recall that $T = 0.04$)

$$HP(w) = HP(z)\Big|_{z=\left(\frac{1+0.02w}{1-0.02w}\right)} = \frac{1-e^{0.08}}{1+e^{0.08}}\frac{(w-50)}{w+50\frac{1-e^{0.08}}{1+e^{0.08}}} \approx -0.04\frac{w-50}{w-2}.$$

[2 marks]

c) The open-loop transfer function corresponding to the cascade of $C(w)$ and $HP(w)$ is

$$C(w)HP(w) = -0.04k\frac{(w-50)(w+a)}{(w-2)(w+b)}.$$

Hence, the closed-loop transfer function is

$$\frac{C(w)HP(w)}{1+C(w)HP(w)} = -0.04k\frac{(w-50)(w+a)}{(w-2)(w+b)-0.04k(w-50)(w+a)}.$$

Consequently, the DC gain is

$$\left.\frac{C(w)HP(w)}{1+C(w)HP(w)}\right|_{w=0} = \frac{ak}{ak-b}.$$

The characteristic polynomial can be written as

$$(1-0.04k)w^2 + (b+2k-0.04ak-2)w - 2b + 2ak.$$

Since we want that this characteristic polynomial is a multiple of $(w+1)(w+2) = w^2+3w+2$, the coefficients k, a and b must be such that

$$\begin{cases} \dfrac{b+2k-0.04ak-2}{1-0.04k} = 3, \\ \dfrac{2ak-2b}{1-0.04k} = 2, \\ \dfrac{ak}{ak-b} = 2, \end{cases}$$

where the last equation imposes a DC gain of 2. This system of equations has a unique solution given by $k = 1.9585$, $a = 0.9412$ and $b = 0.9217$. [8 marks]

d) The discrete-time controller is

$$C(z) = [C(w)]_{w=50\left(\frac{z-1}{z+1}\right)} = 0.6033\frac{z+0.315}{z-0.6033}.$$

The open-loop transfer function is

$$C(z)HP(z) = 0.0503\frac{z+0.315}{(z-1.0833)(z-0.6033)},$$

Answers

hence the closed-loop transfer function is

$$\frac{C(z)HP(z)}{1+C(z)HP(z)} = 0.0503\frac{z+0.315}{(z-1.0833)(z-0.6033)+0.05(z+0.315)}.$$

Consequently, the DC gain is

$$\left.\frac{C(z)HP(z)}{1+C(z)HP(z)}\right|_{z=1} \approx 2.$$

The characteristic polynomial is

$$z^2 - 1.6366z + 0.6693$$

and the closed-loop poles are

$$z_1 = 0.8135, \qquad z_2 = 0.8228.$$

[6 marks]

Answers

2. a) The solid-grey s-plane region is mapped into the shaded z-plane region shown in Figure 2.1. The five points to be mapped are marked by white dots.

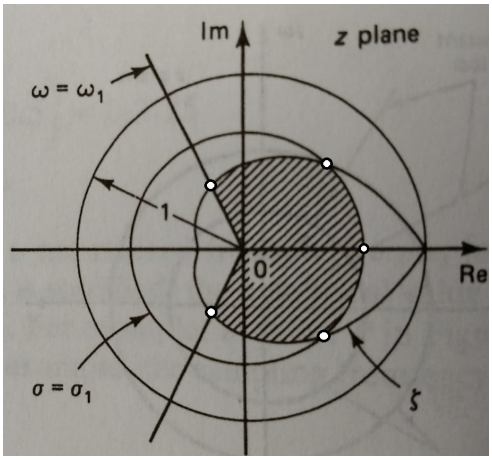


Figure 2.1 Shaded z-plane region.

[6 marks]

b) We introduce (from left to right) the auxiliary variables $e_1(t)$, $e_2(t)$ and $s(t)$ at the entrance of the three unlabeled samplers. At the exit of these three samplers we introduce the variables $e_1^*(t)$, $e_2^*(t)$ and $s^*(t)$. We can now write down the relations between these variables in the Laplace domain, namely

$$\begin{aligned}
E_1(s) &= R(s) - H_1(s)Y(s),\\
E_2(s) &= C_1(s)E_1^*(s) - H_2(s)Y(s),\\
S(s) &= C_2(s)E_2^*(s),\\
Y(s) &= G(s)S^*(s).
\end{aligned}$$

We substitute the last equation in the first two and we determine the *starred* version of the equations

$$\begin{aligned}
E_1^*(s) &= R^*(s) - [H_1(s)G(s)]^*S^*(s),\\
E_2^*(s) &= C_1^*(s)E_1^*(s) - [H_2(s)G(s)]^*S^*(s),\\
S^*(s) &= C_2^*(s)E_2^*(s),\\
Y^*(s) &= G^*(s)S^*(s).
\end{aligned}$$

Substituting the first and third equation in the second, yields

$$E_2^*(s) = C_1^*(s)R^*(s) - C_1^*(s)[H_1(s)G(s)]^*C_2^*(s)E_2^*(s) - [H_2(s)G(s)]^*C_2^*(s)E_2^*(s).$$

From this equation, we solve with respect to $E_2^*(s)$, replace the result in the expression for $Y^*(s)$ and divide by $R^*(s)$, yielding

$$\frac{Y^*(s)}{R^*(s)} = \frac{C_1^*(s)C_2^*(s)G^*(s)}{1 + C_1^*(s)C_2(s)^*(s)\,[H_1(s)G(s)]^* + C_2^*(s)\,[H_2(s)G(s)]^*}.$$

From this last equation we can write directly the pulse transfer function

$$\frac{Y(z)}{R(z)} = \frac{C_1(z)C_2(z)G(z)}{1 + C_1(z)C_2(z)GH_1(z) + C_2(z)GH_2(z)}.$$

where $GH_1(z) = Z\,[G(s)H_1(s)]$ and $GH_2(z) = Z\,[G(s)H_2(s)]$.

[7 marks]

Answers

c) We introduce the auxiliary variables $E_1(z)$ and $E_2(z)$ before the blocks $C_1(z)$ and $C_2(z)$, respectively. We can now write down the relations between $D(z)$, $E_1(z)$, $E_2(z)$ and $Y(z)$, namely

$$\begin{aligned} Y(z) &= D(z) + G(z)C_2(z)E_2(z), \\ E_2(z) &= C_1(z)E_1(z) - H_2(z)Y(z), \\ E_1(z) &= -H_1(z)Y(z). \end{aligned}$$

Substituting the third equation in the second and the resulting second equation in the first yields

$$Y(z) = D(z) - G(z)C_2(z)C_1(z)H_1(z)Y(z) - G(z)C_2(z)H_2(z)Y(z).$$

Solving with respect to $Y(z)$ and dividing by $D(z)$ yields

$$\frac{Y(z)}{D(z)} = \frac{1}{1 + G(z)C_2(z)C_1(z)H_1(z) + G(z)C_2(z)H_2(z)}.$$

The gain of $C_1(z)$ or of $C_1(z)C_2(z)$ should be selected as large as possible (*i.e.* satisfying other specifications, such as stability) to minimize the effect of the disturbance on the output. [7 marks]

Answers

3. a) The equivalent discrete-time model is

$$\begin{aligned} HP(z) &= (1-z^{-1})Z\left(\frac{e^{-2Ts}}{s(s+2)}\right) = z^{-2}(1-z^{-1})Z\left(\frac{1}{s(s+2)}\right) \\ &= \frac{1-e^{-T}}{z^3-e^{-T}z^2} = \frac{1}{2z^3-z^2}. \end{aligned}$$

[4 marks]

b) The characteristic polynomial is given by

$$p(z) = 2z^3 - z^2 + k.$$

Apply the transformation $z = \dfrac{1+w}{1-w}$ to obtain

$$p(w) = p(z)|_{z=\left(\frac{1+w}{1-w}\right)} = \frac{(3-k)w^3 + (7+3k)w^2 + (5-3k)w + 1 + k}{(w-1)^3}.$$

Then we apply the Routh criterion on

$$(3-k)w^3 + (7+3k)w^2 + (5-3k)w + 1 + k.$$

The Routh table is

$$\begin{array}{ll} 3-k & 5-3k \\ 7+3k & 1+k \\ \dfrac{8(K^2+k-4)}{7+3k} & \\ 1+k & \end{array}$$

Imposing that each element of the first column of the table is positive, yields that the closed-loop system is asymptotically stable for

$$(k_{\min}, k^{\max}) = \left(0, \frac{1}{2}(-1+\sqrt{17})\right) \approx (0, 1.5616).$$

[6 marks]

c) The root locus is displayed in Figure 3.1.

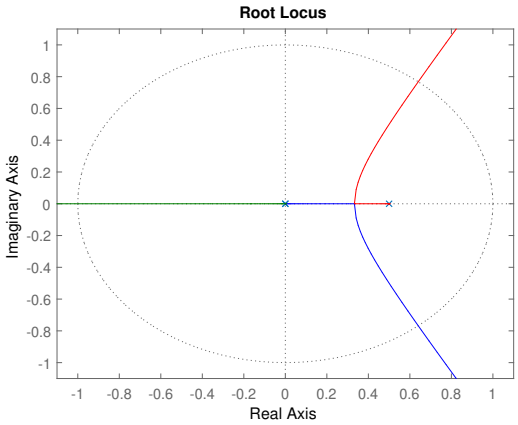


Figure 3.1 Positive root locus.

The locus has three open-loop poles, one at $\dfrac{1}{2}$ and two at 0, and three open-loop zeros at infinity. The locus on the real axis lies on $(-\infty, 0) \cap (0, \dfrac{1}{2})$. We compute

Answers

the asymptote intersection, which yields $\sigma = \frac{\frac{1}{2}}{3} = \frac{1}{6}$ and the departure angles, which yields $\theta_1 = -\frac{\pi}{3}$, $\theta_2 = \pi$, $\theta_3 = \frac{\pi}{3}$. We compute the break-in/break-out points by solving

$$6z^2 - 2z = 0$$

which yields $z_1 = 0$ and $z_2 = \frac{1}{3}$. The point z_1, corresponding to $k = 0$, is the break-away point of the negative locus, which we ignore. Thus we have found one break-away point for the positive locus, namely $z_2 = \frac{1}{3}$. The locus departs from the open-loop poles and approached the three open-loop zeros at infinity. There are three points of exit of the locus from the unit circle. At least one of these points corresponds to the gain $k = k_{\max}$. In fact, for $k = k_{\max}$, the closed-loop poles are -0.7808 and $0.6404 \pm 0.7681i$ (the modulus of which is 0.7808 and 1, respectively). [6 marks]

d) Define the change of variable $x = \frac{1}{\alpha} z$ and note that

$$|x| = \left|\frac{1}{\alpha} z\right| = \frac{1}{\alpha}|z|.$$

Hence, if $|x| < 1$, then $1 > \frac{1}{\alpha}|z|$ and $|z| < \alpha$. Thus, we can solve this question by using the Routh criterion on the characteristic polynomial obtained using the transformation $x = \frac{1+w}{1-w}$ on

$$HP(x) = HP(z)\Big|_{x = \frac{1}{\alpha} z}.$$

[4 marks]

Answers

4. a) As hinted, substituting $t = (k+1)T$ and $t_0 = kT$ in

$$x(t) = e^{a(t-t_0)}x(t_0) + \int_{t_0}^{t} e^{a(t-\tau)}bu(\tau)d\tau, \tag{4.1}$$

yields

$$x((k+1)T) = e^{aT}x(kT) + \int_{kT}^{(k+1)T} e^{a((k+1)T-\tau)}bu(\tau)d\tau.$$

Notice that in the interval $[kT,(k+1)T)$, $u(t)$ is identically equal to $u(kT)$. Thus, $u(kT)$ can be brought outside the integral. Again as hinted, using the change of variable $\lambda = (k+1)T - \tau$ yields

$$x((k+1)T) = e^{aT}x(kT) + \int_{T}^{0} e^{a\lambda}b(-d\lambda)u(kT),$$

from which the relations

$$f = e^{aT}, \qquad g = \int_0^T e^{a\lambda}bd\lambda = \frac{b}{a}\left(e^{aT}-1\right),$$

follow.

[2 marks]

b) Substituting $t_0 = kT$ in (4.1) yields

$$x(t) = e^{a(t-kT)}x(kT) + \int_{kT}^{t} e^{a(t-\tau)}bu(\tau)d\tau = \xi(t-kT)x(kT) + \eta(t-kT)u(kT)$$

where

$$\xi(t-kT) = e^{a(t-kT)}, \qquad \eta(t-kT) = \int_{kT}^{t}\xi(t-\tau)bd\tau = \frac{b}{a}\left(e^{a(t-kT)}-1\right).$$

[6 marks]

c) Substituting

$$x(t) = \xi(t-kT)x(kT) + \eta(t-kT)u(kT)$$

in

$$J = \frac{1}{2}x^2(NT) + \frac{1}{2}\sum_{k=0}^{N-1}\int_{kT}^{(k+1)T}[qx(t)^2 + ru(t)^2]d\tau$$

yields

$$\begin{aligned}
J &= \frac{1}{2}x^2(NT) + \frac{1}{2}\sum_{k=0}^{N-1}\int_{kT}^{(k+1)T}\{q[\xi(t-kT)x(kT) + \eta(t-kT)u(kT)]^2 + ru(kT)^2\}d\tau \\
&= \frac{1}{2}x^2(NT) + \frac{1}{2}\sum_{k=0}^{N-1}\int_{kT}^{(k+1)T}[q\xi(t-kT)^2x(kT)^2 + q\eta(t-kT)^2u(kT)^2 \\
&\quad +2q\xi(t-kT)\eta(t-kT)x(kT)u(kT) + ru(kT)^2]d\tau \\
&= \frac{1}{2}x^2(NT) + \frac{1}{2}\sum_{k=0}^{N-1}[\widetilde{q}x(kT)^2 + 2\widetilde{m}x(kT)u(kT) + \widetilde{r}u(kT)^2]d\tau,
\end{aligned}$$

where

$$\begin{aligned}
\widetilde{q} &= \int_{kT}^{(k+1)T} q\xi(\tau-kT)^2d\tau = \frac{q}{2a}(e^{2aT}-1), \\
\widetilde{m} &= \int_{kT}^{(k+1)T} q\xi(\tau-kT)\eta(\tau-kT)d\tau = \frac{bq}{2a^2}(e^{aT}-1)^2, \\
\widetilde{r} &= \int_{kT}^{(k+1)T}[q\eta(\tau-kT)^2 + r]d\tau = \frac{b^2q}{2a^3}[(e^{aT}-3)(e^{aT}-1)+2aT] + rT.
\end{aligned}$$

[8 marks]

Answers

d) i) The obtained performance index contains a cross-term involving $x(kT)$ and $u(kT)$, whereas the original continuous-time performance index does not have a cross-term. This difference is due to the discretization of $x(t)$, which contains the sum of terms in $x(kT)$ and $u(kT)$. Then the cross-term arises by squaring this sum in the performance index.

ii) In class we have seen that the discrete-time LQR with a cross-term involving $x(kT)$ and $u(kT)$ can be rewritten in a discrete-time LQR problem without cross-term (provided some positive definiteness condition holds) by defining a new q and a new controller v.

[4 marks]

Answers